MakersSoc Tinkercad QuickStart Guide

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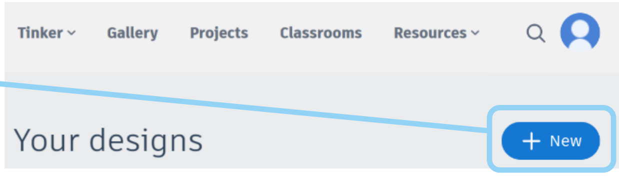
TinkerCAD is a free web app for 3D modelling, simulating electronics and their programming.

3D modelling

1. Create an account & login
   Go to www.tinkercad.com.
   An Autodesk login can be used. If you have a login for Fusion 360, that can be used here too.

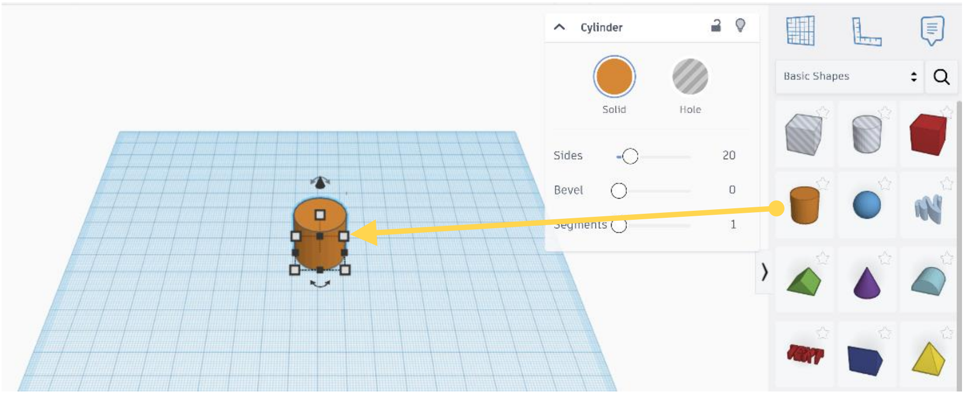
2. Create a new project
   "+ New" --> 3D design



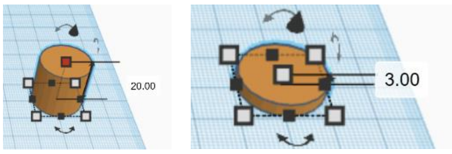
   Name it something memorable.
   Maybe add some version control to the naming scheme. "MakersSoc Keyring V1" (or "2022-09-26 09-46 MakersSoc Keyring i1")

3. Add a shape
   Drag a shape from the left hand side.

   For the keyring, use a cylinder.

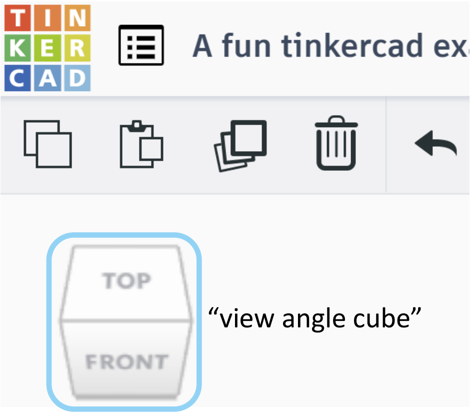


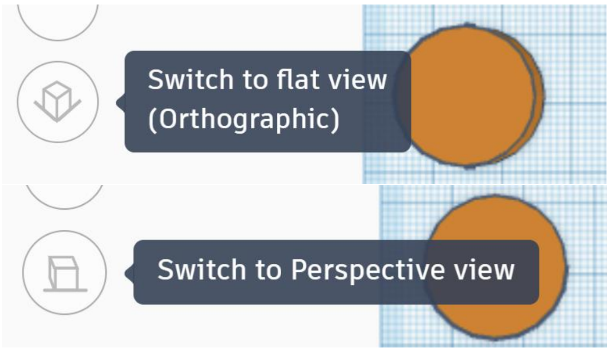
   Increase segments to 10.
   Change height to 3mm.
   The default unit is millimetres.



   At any point, actions can be undone with the undo button or shortcut (crtl+z).

4. <u>Adjusting camera view</u>
Using the view angle cube (in the top right corner) or crtl+click & dragging in space, the view is adjusted for a top down perspective.





Change the view from perspective to orthographical view using the toggle on the left hand side.

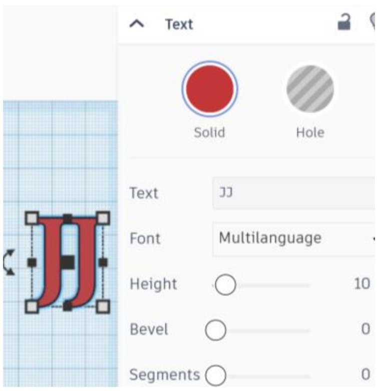
In the perspective view (the default), objects which are far away are smaller than those nearby. In the orthographic view, all objects appear at the same scale. This can be useful for aligning elements among other factors when designing.

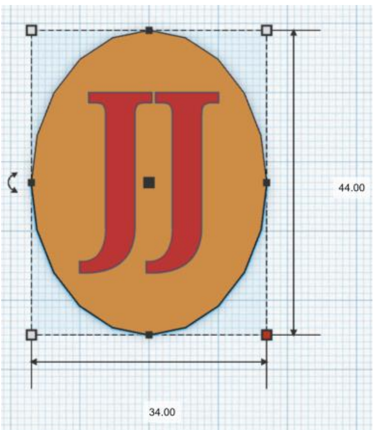
5. Add your name as text

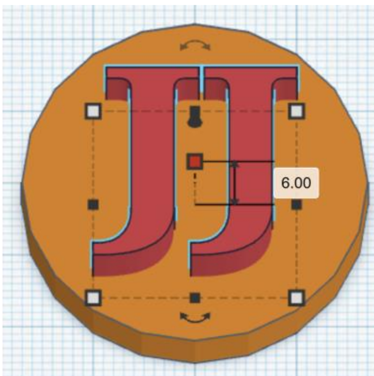
6. Adjust the size of circle to fit the text with plenty of room.
Also adjusted name dimensions.
Adjusting it to dimensions that are easy to centre. To centre the text, align the circles & text's left side, take the width of the circle away from text's width and then divide by two. Move the text to the right by this value. Then do the same with the depth.

$$\frac{circle - text}{2}$$

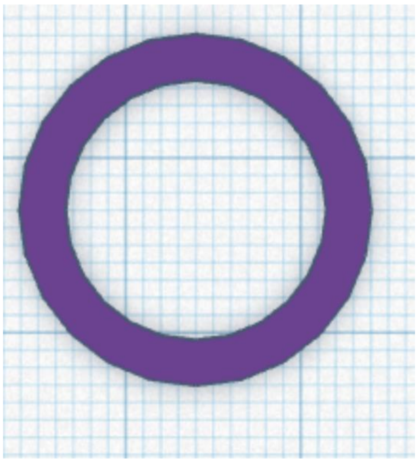
Set the height of text to 6mm

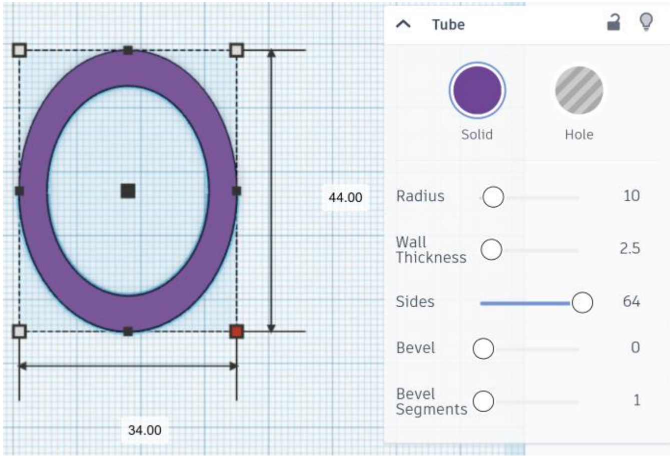






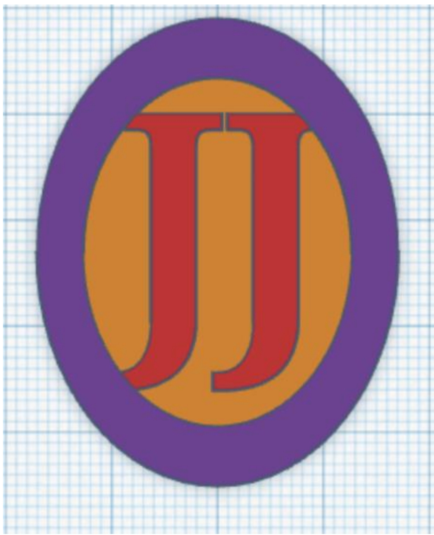
7. Insert a new ring/tube element

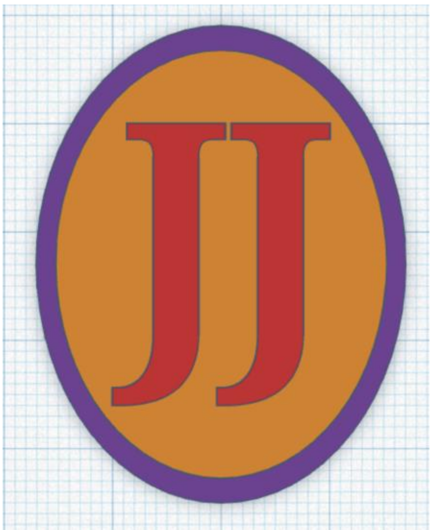




Change dimensions to match / to be the same as the cylinder.
Set sides to 64.

Aligned with remainder of project / with cylinder.
Adjust the wall thickness of the ring until it was appropriate. If unsure, get a member of the committee to check.

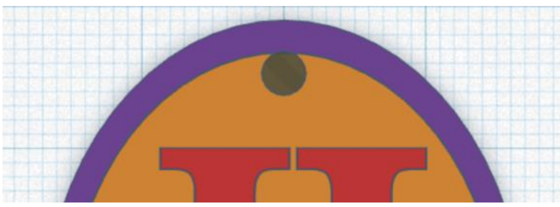




8. Add a hole element

Set its dimensions to be 3x3mm

Align with rest of the project.



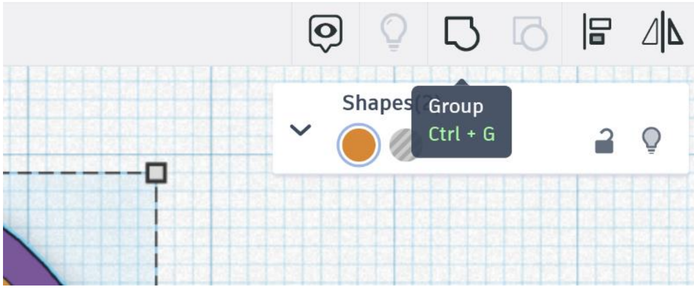
9. Merging

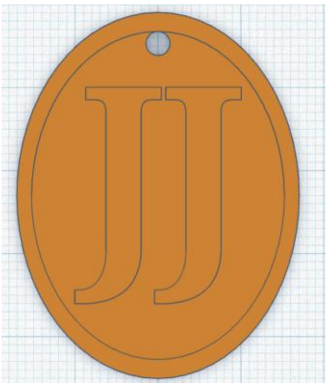
Set the outline's height to 6mm.

Merge the base cylinder and the outline ring by selecting both of them – doing this first means if the hole does overlap with the either, it will cut into both (as the merged object).
Multiple elements can be selected by holding shift. Merging can also be done using ctrl+g with selected objects.

Merge the text and the body of the keyring.

Merge the hole with the keyring.

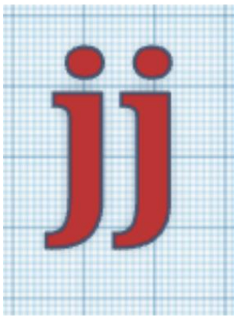




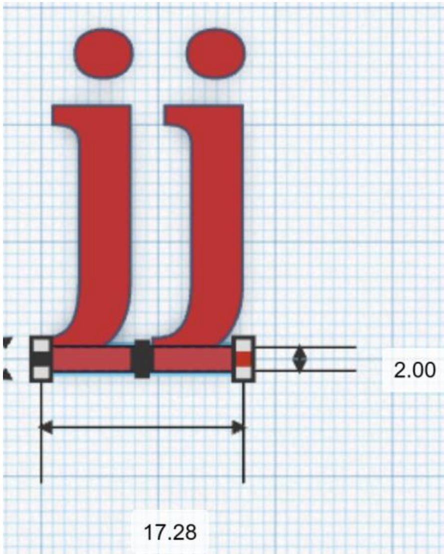
10. Export as STL.
The file is now ready to be sent off to be 3D printed.
STL files are used for 3D printing.

Laser Cutting

1. Create and name a new project
2. Insert text and write a word like your name or initials. To keep cutting time low, please keep the project within a 2x5cm region. This can be measured with the ruler, found under the export button.
3. Switch to orthographical view and view project from above.



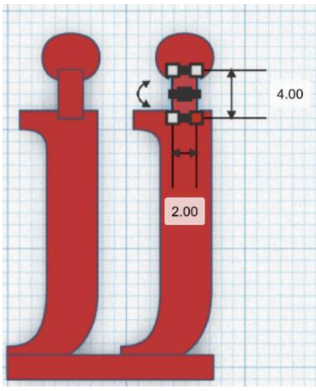
4. Insert a cube and adjust its dimensions so it joins each letter.



5. Some letters have parts separate from the main part of the letter. These parts will need to be connected to the rest of the text.

   Join the additional parts of letters with cubes.

   "i"s, "j"s, accents, exclamation marks, question marks, colons, percentage signs are several examples of these kind of characters.



6. Merge all elements together. Make sure there are no gaps as otherwise they will be cut as separate pieces.

7. Export as an SVG file. SVG files are used for laser cutting.